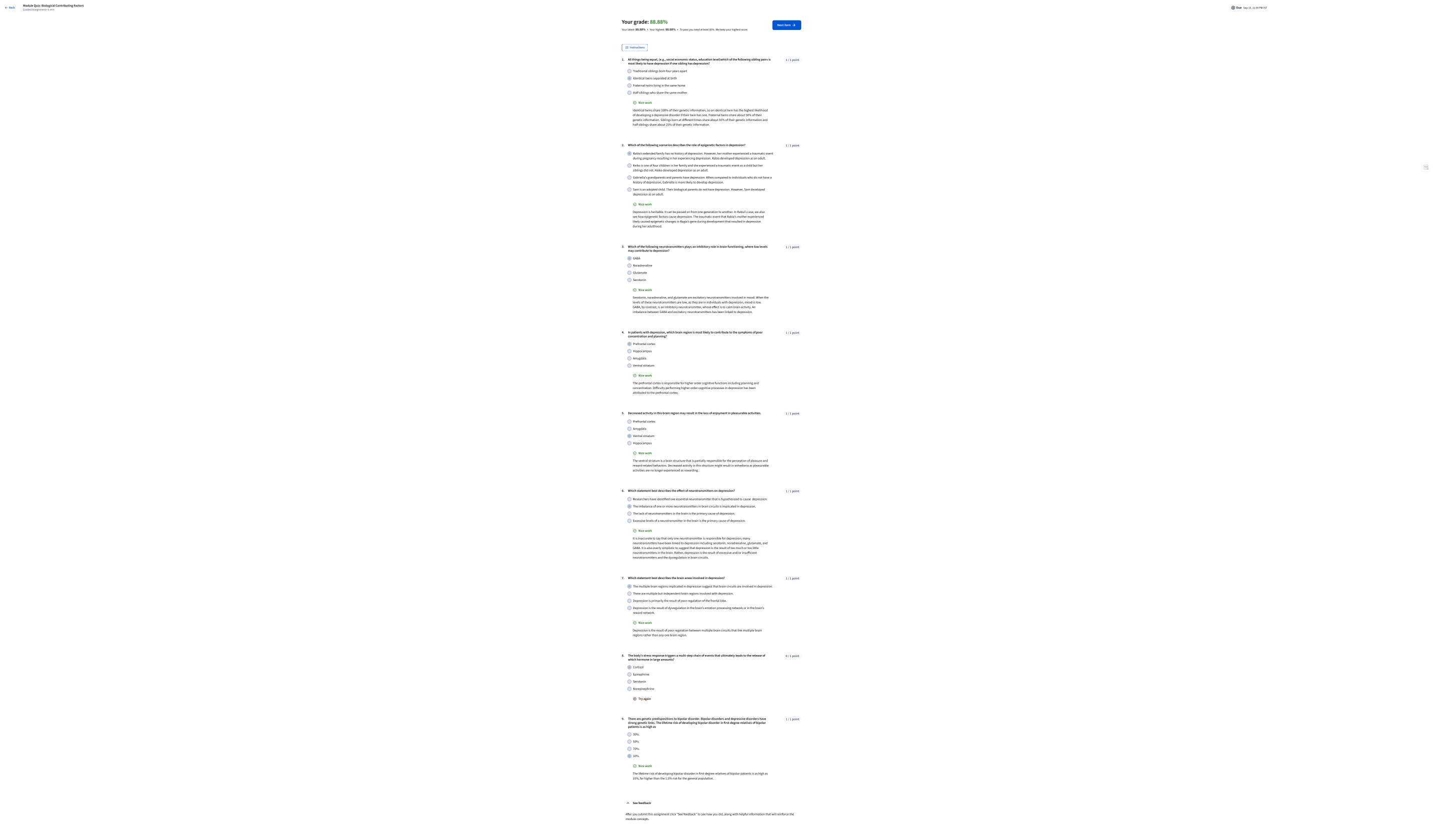

## Module Quiz: Biological Contributing Factors

Graded Assignment • 9 min

**Due** Sep 15, 11:59 PM IST

## Your grade: 88.88%

Your latest: **88.88%** • Your highest: **88.88%** • To pass you need at least 80%. We keep your highest score.

1. **All things being equal, (e.g., social economic status, education level)which of the following sibling pairs is most likely to have depression if one sibling has depression?** 1 / 1 point

○ Traditional siblings born four years apart
● Identical twins separated at birth
○ Fraternal twins living in the same home
○ Half siblings who share the same mother

✓ Nice work

Identical twins share 100% of their genetic information, so an identical twin has the highest likelihood of developing a depressive disorder if their twin has one. Fraternal twins share about 50% of their genetic information. Siblings born at different times share about 50% of their genetic information and half siblings share about 25% of their genetic information.

2. **Which of the following scenarios describes the role of epigenetic factors in depression?** 1 / 1 point

● Rabia's extended family has no history of depression. However, her mother experienced a traumatic event during pregnancy resulting in her experiencing depression. Rabia developed depression as an adult.
○ Keiko is one of four children in her family and she experienced a traumatic event as a child but her siblings did not. Keiko developed depression as an adult.
○ Gabriella's grandparents and parents have depression. When compared to individuals who do not have a history of depression, Gabriella is more likely to develop depression.
○ Sam is an adopted child. Their biological parents do not have depression. However, Sam developed depression as an adult.

✓ Nice work

Depression is heritable. It can be passed on from one generation to another. In Rabia's case, we also see how epigenetic factors cause depression. The traumatic event that Rabia's mother experienced likely caused epigenetic changes in Ragia's gene during development that resulted in depression during her adulthood.

3. **Which of the following neurotransmitters plays an inhibitory role in brain functioning, where low levels may contribute to depression?** 1 / 1 point

● GABA
○ Noradrenaline
○ Glutamate
○ Serotonin

✓ Nice work

Serotonin, noradrenaline, and glutamate are excitatory neurotransmitters involved in mood. When the levels of these neurotransmitters are low, as they are in individuals with depression, mood is low. GABA, by contrast, is an inhibitory neurotransmitter, whose effect is to calm brain activity. An imbalance between GABA and excitatory neurotransmitters has been linked to depression.

4. **In patients with depression, which brain region is most likely to contribute to the symptoms of poor concentration and planning?** 1 / 1 point

● Prefrontal cortex
○ Hippocampus
○ Amygdala
○ Ventral striatum

✓ Nice work

The prefrontal cortex is responsible for higher order cognitive functions including planning and concentration. Difficulty performing higher order cognitive processes in depression has been attributed to the prefrontal cortex.

5. **Decreased activity in this brain region may result in the loss of enjoyment in pleasurable activities.** 1 / 1 point

○ Prefrontal cortex
○ Amygdala
● Ventral striatum
○ Hippocampus

✓ Nice work

The ventral striatum is a brain structure that is partially responsible for the perception of pleasure and reward-related behaviors. Decreased activity in this structure might result in anhedonia as pleasurable activities are no longer experienced as rewarding.

6. **Which statement best describes the effect of neurotransmitters on depression?** 1 / 1 point

○ Researchers have identified one essential neurotransmitter that is hypothesized to cause depression.
● The imbalance of one or more neurotransmitters in brain circuits is implicated in depression.
○ The lack of neurotransmitters in the brain is the primary cause of depression.
○ Excessive levels of a neurotransmitter in the brain is the primary cause of depression.

✓ Nice work

It is inaccurate to say that only one neurotransmitter is responsible for depression; many neurotransmitters have been linked to depression including serotonin, noradrenaline, glutamate, and GABA. It is also overly simplistic to suggest that depression is the result of too much or too little neurotransmitters in the brain. Rather, depression is the result of excessive and/or insufficient neurotransmitters and the dysregulation in brain circuits.

7. **Which statement best describes the brain areas involved in depression?** 1 / 1 point

● The multiple brain regions implicated in depression suggest that brain circuits are involved in depression.
○ There are multiple but independent brain regions involved with depression.
○ Depression is primarily the result of poor regulation of the frontal lobe.
○ Depression is the result of dysregulation in the brain's emotion processing network or in the brain's reward network.

✓ Nice work

Depression is the result of poor regulation between multiple brain circuits that link multiple brain regions rather than any one brain region.

8. **The body's stress response triggers a multi-step chain of events that ultimately leads to the release of which hormone in large amounts?** 0 / 1 point

● Cortisol
○ Epinephrine
○ Serotonin
○ Norepinephrine

⊗ Try again

9. **There are genetic predispositions to bipolar disorder. Bipolar disorders and depressive disorders have strong genetic links. The lifetime risk of developing bipolar disorder in first-degree relatives of bipolar patients is as high as** 1 / 1 point

○ 30%.
○ 50%.
○ 70%.
● 10%.

✓ Nice work

The lifetime risk of developing bipolar disorder in first-degree relatives of bipolar patients is as high as 10%, far higher than the 1.5% risk for the general population.

**See feedback**

After you submit this assignment click "See feedback" to see how you did, along with helpful information that will reinforce the module concepts.